Teacher only; do not distribute D or the complete pack to students

Teaching arrangements and study conditions

1. The experimental group uses IBL + VL. The comparison group has no pre-lesson lecture and does not operate the VL; it learns through the WS. Make A available from the start: explanation → example → guided practice → independent application → checking/correction.
2. Both groups use 15–20 minutes, preferably the same 20-minute limit: concepts 3, examples 2, guided practice 4, independent application 8, checking/correction 3 minutes. If using 15 minutes, use the same limit for both groups; speed is not an ability score.
3. Teachers may briefly clarify terms, diagram-reading instructions and response requirements already written in the WS, without a separate concept lecture before self-study. Examples do not use X/Y/Z/C outcomes. Do not reveal B answers before completion. Record additional prompts and actual completion times.
4. B1–2 are guided; B3–5 contain twelve independent measurements; B6 covers means and six analysis categories; B7 applies resistance（抗藥性） selection. Accept short sentences or key terms. Two lines provide space; they need not be filled.
5. After B is completed, preserve the first answers before distributing C for corrections. Only then provide answers and focused feedback. Do not give answers to unfinished students. Record unfinished items and pilot feasibility of 15–20 minutes; software tests do not establish actual learning duration.
6. All diagrams and rulers use existing fixed seeds 1001, 1002 and 1003 with generateResult. Individual VL seeds differ; do not claim all learner values are identical. 30°C and 24 hours are simulation conditions. No physical MRSA（耐甲氧西林金黃葡萄球菌） culture is required.
7. Check external-zone measurements within ±1 mm of the model. Absence requires No and exactly 0. Check means against the student’s own readings (±0.1 mm), avoiding a second penalty for one measurement error.
8. The 38-mark WS check is specific to this task version, not directly comparable with the VL’s 23 marks; paper ordering/reading is not practical performance. Use common SPSAI/CKT assessments for study outcomes; researchers must verify the formal instruments and their validity.
9. This redesign follows the researcher’s 14 October 2026 comparison condition, replacing the earlier same-item, inquiry-before-teaching arrangement. Use the alignment table to update study documents. Preserve approved bilingual vocabulary; do not change learner VL, records, cloud collection or Excel.

Ordering and diagram reading do not demonstrate virtual practical skill. WS marks are not equivalent to VL prediction, operation or completion scores. Evaluate group outcomes using separate common assessments.

Teacher only; do not distribute D or the complete pack to students

Check your measurements and analysis

Distribute only after B is completed and first answers preserved. Check, then correct in another colour without erasing the original.

Plate	Sample	Visible zone outside the disc?	Total clear-zone（清晰區） diameter (mm)
1	X	Yes	18.7
	Y	No	0.0
	Z	Yes	26.1
	C (control)	No	0.0
2	X	Yes	18.7
	Y	No	0.0
	Z	Yes	26.3
	C (control)	No	0.0
3	X	Yes	18.7
	Y	No	0.0
	Z	Yes	26.5
	C (control)	No	0.0

Mean total diameter calculation	X	Y	Z	C (control)
mm	18.7	0.0	26.3	0.0

Reference values are not a reason to overwrite valid readings. Ruler readings may differ within ±1 mm; check means against your own readings.

1. What does the appearance of a clear zone（清晰區） indicate?	B. Under these test conditions, there is no visible bacterial growth in the zone, indicating inhibition（抑制）.
2. What does the diameter of a clear zone（清晰區） represent?	C. The extent of the area with no visible bacterial growth under these conditions; diffusion and other factors also affect the diameter.
4. Why do experiments need to be repeated?	C. To reduce random error and assess the reproducibility（可重複性） of the results and the reliability（可靠性） of the data.
5. If repeated results differ greatly, what should you do, and why?	D. First check spreading, disc positions and plate conditions, then repeat as needed to identify causes of variation and improve the comparison.
6. When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?	C. Provide an antibiotic（抗生素）-free comparison to help assess whether clear-zone（清晰區） formation is associated with the antibiotic（抗生素） sample.

3. Ranking of mean total clear-zone（清晰區） diameter under these conditions: Z ＞ X ＞ Not applicable ＞ Not applicable。

Samples with external zones: X、Z。

Teacher only; do not distribute D or the complete pack to students

Check reasoning and correct

Distribute only after B is completed and first answers preserved. Check, then correct in another colour without erasing the original.

Your initial observations

A clear zone（清晰區） with no visible bacterial growth is present near Penicillium（青黴菌）(a fungus). This is an observation; its cause needs testing.

Independent variable	Sample in the disc
Dependent variable	Visible clear zone（清晰區） and its total diameter
Controlled variables	Disc size ／ Composition of the agar plate（瓊脂板） ／ Type of bacteria used ／ Initial bacterial distribution ／ Preset sample preparation
What are the assumptions of this investigation?	Bacteria are initially distributed evenly on each agar plate（瓊脂板）. ／ Equipment does not introduce contamination. ／ All plates use the same simulated observation conditions.

Use a matching carrier-only disc without antibiotic（抗生素） to check carrier/disc effects, with other conditions the same. Call it a water control only if every carrier is water.

Hypothesis evaluation/application

B is testable because it compares a sample/carrier under shared conditions using zone presence and diameter. Accept a reasoned conditional hypothesis for U and another bacterium, without requiring an outcome guess.

Labels, spacing and errors

Use X/Y/Z/C once each, in any order. Space discs apart and away from the edge to avoid overlap and difficult interpretation. A risks contamination; B changes bacteria and makes the comparison unfair; C is appropriate.

Repeat-plan comparison

Plan B uses new independently prepared plates to compare variation. Remeasuring one plate estimates reading variation, not independent experimental variation. Three is a classroom choice; balance resources, time and useful evidence.

Procedure and errors

Order: R → U → Q → T → S → P. A: check even spreading. B: prepare a new independent plate rather than call a second reading a new result. Retain differing data and investigate causes.

Concept application

For example, X has readings 18.7, 18.7, 18.7 mm and an external zone. Pre-existing resistant（具抗藥性的） bacteria are more likely to survive and reproduce（繁殖）; traits can be inherited（遺傳）. Their proportion may rise, so effectiveness may fall without a complete takeover. Accept Z with its own readings. The experiment does not prove resistance（抗藥性） arose.

My correction and reason (if originally correct, state the check used)

Correct the evidence or causal link in your concept application.

Teacher only; do not distribute D or the complete pack to students

WS task check: 38 marks (not a cross-group outcome score)

Ordering and diagram reading do not demonstrate virtual practical skill. WS marks are not equivalent to VL prediction, operation or completion scores. Evaluate group outcomes using separate common assessments.

Item	Marks	Criterion
Initial observation	2	Describe reduced visible growth/clear area near the fungus and separate observation from causal inference; 1 each.
Hypothesis evaluation	1	Select B and identify comparison conditions and an observable outcome; 1 for a reasoned judgment.
Hypothesis in another context	2	Testable comparison of U/carrier for another bacterium: 1; observable zone/diameter outcome: 1. Do not score outcome guessing.
Identify the variables	4	Independent: 1; dependent: 1; five correct controls: 0.4 each, with 0 for inappropriate extra control choices.
What are the assumptions of this investigation?	1	Only even distribution, no added contamination and shared conditions; missing/additional choices: 0.
Control explanation	2	Same carrier without antibiotic（抗生素）: 1; compare carrier/other factors under shared conditions: 1. Not necessarily a water control.
Complete labels and check fair comparison	4	Four labels once each: 1; spacing/edge reasoning: 1; identify A contamination and B unfair comparison: 1 each. No art or operation score.

Teacher only; do not distribute D or the complete pack to students

WS task check: 38 marks (not a cross-group outcome score)

Ordering and diagram reading do not demonstrate virtual practical skill. WS marks are not equivalent to VL prediction, operation or completion scores. Evaluate group outcomes using separate common assessments.

Item	Marks	Criterion
Independent repeats and feasibility	3	Plan B and independent new plates: 1; variation comparison: 1; three is not universal, with time/resource feasibility: 1.
Diagram reading and procedure ordering (not practical performance)	4	R→U→Q→T→S→P: 2 (1/3 per correct position); A spreading problem and B non-independent result: 1 each. Procedure understanding only.
Measure and record	4	Twelve readings at 1/3 each: correct zone classification plus ±1 mm; absence requires No and exactly 0.
Mean total diameter calculation	2	Four means at 0.5 each, based on the student’ s own three readings, within ±0.1 mm.
Analyse the results and draw a conclusion	6	Five MCs at 1 each: B, C, C, D, C. Item 3: samples 0.5, ranking by own means 0.5.
Resistance（抗藥性）- selection application	3	Effective sample/readings: 1; differential survival, reproduction（繁殖） and inheritance（遺傳）: 1; proportion may rise/effectiveness may decline: 1. No complete takeover required.

Teacher only; do not distribute D or the complete pack to students

Item alignment: original, treatment, new item, shared goal and rationale

VL original	Treatment	New WS item	Shared goal	Rationale
Your initial observations	Retain / static image	Refer to the historical image in A. Describe visible bacterial growth near the fungus, separating observation from an explanation of its cause.	Distinguish observation from inference	Retain historical evidence without VL operation.
Which samples do you predict will have an area without visible bacterial growth around the disc?	Rewrite: evaluate a hypothesis	Compare two claims: A, “This sample must kill all MRSA（耐甲氧西林金黃葡萄球菌）”; B, “Under the same conditions, the total clear-zone（清晰區） diameter around a disc with sample M is greater than around the carrier control.” Circle the more testable claim (A/B), and explain why clear-zone（清晰區） data can test it.	Testable hypothesis	Concepts are taught first; do not pretend to make an initial prediction.
Which sample do you predict will have the largest clear zone（清晰區） around its disc?	Rewrite: another context	Another context: compare unknown sample U with the same carrier control using a different bacterium. Write a hypothesis testable with external-zone presence and total diameter; do not make a new prediction about X/Y/Z.	Transfer testable hypotheses	Use unknown U and another bacterium instead of re-predicting seen outcomes.
Your reason (Hint: antibiotics（抗生素） can diffuse through the agar plate（瓊脂板）.)	Integrate with evaluation/application	Compare two claims: A, “This sample must kill all MRSA（耐甲氧西林金黃葡萄球菌）”; B, “Under the same conditions, the total clear-zone（清晰區） diameter around a disc with sample M is greater than around the carrier control.” Circle the more testable claim (A/B), and explain why clear-zone（清晰區） data can test it.	Explain using observable evidence	Evaluate testability and retain reasoning practice.
Independent variable	Retain, with an earlier example	Independent variable	Identify the changed factor	Use a disc-size example, then classify sample contents in X/Y/Z.
Dependent variable	Retain	Dependent variable	Identify observed/measured outcomes	Still classify zone presence and total diameter.
Controlled variables	Retain	Controlled variables	Fair-comparison conditions	Retain the same options and control goals.
What are the assumptions of this investigation?	Retain with definition	Investigation assumptions (choose all that apply)	Identify assumptions	Do not assume absence of visible growth proves complete killing.

Teacher only; do not distribute D or the complete pack to students

Item alignment: original, treatment, new item, shared goal and rationale

VL original	Treatment	New WS item	Shared goal	Rationale
If you want to find out whether antibiotics（抗生素） affect bacterial growth, how would you design a control group?	Rewrite: explain the given control	What is C? Explain its comparison baseline and why the remaining conditions must be the same.	Control comparison function	C has the same carrier without antibiotic（抗生素）, not necessarily water.
Draw and label your plates ／ Setup labels / written design	Rewrite: labels, spacing and error identification	Label the four discs X, Y, Z and C, once each. Explain why discs need space between them and from the edge. Label order does not affect credit.	Setup and fair comparison	Retain position and contamination judgments without autonomous operation.
There are 4 test samples: X, Y, Z and the control. How many tests do you think each sample needs in total? (Include the first test.)	Rewrite: compare given plans	Plan A: remeasure the same plate three times. Plan B: independently prepare and observe three new plates once each. Which provides independent repeats, and how do they help check variation?	Independent repeats and variation	Distinguish new plates from remeasuring one zone.
Why would you plan it this way?	Rewrite: feasibility reasoning	This lesson uses three plates. Why is “three” not the only correct number for every study? What feasibility limits matter when comparing three with more independent plates?	Evidence quality and feasibility	Three is a classroom arrangement, not a universal standard.
Experiment and record ／ Procedure	Static: before/after images, ordering and errors	These cards describe a simulated procedure. Order the letters logically; prepare three plates independently before the shared observation. Answering does not demonstrate sterilisation（滅菌／消毒）, spreading or incubation（培養） performance.	Procedure understanding and evidence interpretation	Not evidence of sterilisation（滅菌／消毒）, spreading or incubation（培養） performance.
Visible zone outside the disc? ／ Measure and record the total clear-zone（清晰區） diameter (mm).	Retain: twelve static rulers	Measure and record the total clear-zone（清晰區） diameter (mm).	Measure and classify	Use the same seeds/model; do not preprint B answers.
Mean total diameter calculation	Retain with a separate worked example	Mean total diameter calculation	Calculate from own readings	Four three-reading means; avoid repeated measurement penalties.
What does the appearance of a clear zone（清晰區） indicate?	Same item	What does the appearance of a clear zone（清晰區） indicate?	Meaning of clear-zone（清晰區） evidence	Independent interpretation after teaching; no complete-killing inference.

Teacher only; do not distribute D or the complete pack to students

Item alignment: original, treatment, new item, shared goal and rationale

VL original	Treatment	New WS item	Shared goal	Rationale
What does the diameter of a clear zone（清晰區） represent?	Same item	What does the diameter of a clear zone（清晰區） represent?	Diameter meaning and limits	Do not infer patient treatment from zone size.
Draw a conclusion from your results: ／ Comparing the mean total clear-zone （清晰區） diameters in this test, their inhibitory（抑制） effects rank as follows:	Equivalent rewrite: mean-diameter order	Ranking of mean total clear-zone（清晰區） diameter under these conditions:	Draw bounded conclusions from data	Retain sample selection without calling diameter clinical effectiveness.
Why do experiments need to be repeated?	Same item	Why do experiments need to be repeated?	Purpose of repeats	Retain random error, reproducibility（可重複性） and reliability（可靠性）.
If repeated results differ greatly, what should you do, and why?	Same item	If repeated results differ greatly, what should you do, and why?	Evaluate variation and methods	Check causes; do not delete data for disagreement.
When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?	Same item	When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?	Role of the control	Retain comparison against non-antibiotic（抗生素） factors.
Choose an antibiotic（抗生素） sample from your experimental results that can inhibit（抑制） bacterial growth, and cite readings to support your judgement. Suppose this bacterial population initially contains a small number of bacteria resistant（具抗藥性的） to that antibiotic（抗生素）. How might its effectiveness change with continued use? Use the learning points to explain how bacterial survival, reproduction（繁殖） and inheritance（遺傳） of resistance traits（抗藥性特徵） cause this change.	Equivalent concise concept application	Choose a growth-inhibiting（抑制） sample and cite readings. If a few bacteria already resist that antibiotic（抗生素）, how might continued use affect its effectiveness? Explain through survival, reproduction（繁殖）, inheritance（遺傳） and changing proportions.	Data, survival, reproduction（繁殖） and inheritance（遺傳）	The original already applies resistance（抗藥性） selection; no hypothesis-revision task is needed.
Inquiry submitted ／ Learning points	Reorder: concepts in A, checking in C	Distribute only after B is completed and first answers preserved. Check, then correct in another colour without erasing the original.	Self-study and correction	Teach before practice; preserve first answers and do not score completion as ability.